

PRACTICAL TUTORIAL

Applying Cryptography in a University Environment

Module 21: Cryptography

Student Name	
Student Number	
Module	
Date	

Scenario

You have recently joined the University ICT Security Team as a junior cybersecurity analyst. The University processes sensitive information such as student records, examination documents, assessment results, staff information, research data, financial information and institutional communications. Your task is to demonstrate how cryptography can be used to protect this information.

1. Learning Outcomes

- Explain the role of cryptography in protecting University information.
- Use SHA-256 to verify file integrity.
- Encrypt and decrypt sensitive information using symmetric encryption.
- Generate public and private cryptographic keys.
- Use public-key cryptography to protect information.
- Create and verify digital signatures.
- Detect modification of digitally signed information.
- Examine a digital certificate and explain the role of Certificate Authorities and PKI.
- Recommend appropriate cryptographic controls for common University security scenarios.

2. Requirements

- A Linux, macOS or Windows computer.
- OpenSSL installed.
- A terminal or command prompt.
- A text editor.

Check that OpenSSL is available:

```
openssl version
```

3. Create Your Practical Environment

```
mkdir university-crypto-lab
cd university-crypto-lab
pwd
```

PART A - Protecting the Integrity of Examination Documents

Scenario: The Department of Computer Science has prepared an examination document that must be transferred to the Examinations Office. The office needs a way to determine whether the document has been modified.

Activity 1: Create the Examination File

```
cat > examination.txt << 'EOF'
C4CPS505 Examination
Date: 20 November 2026
Time: 09:00
Duration: 3 Hours
Venue: HP Laboratory 1
EOF
cat examination.txt
```

Activity 2: Generate a Digital Fingerprint

```
openssl dgst -sha256 examination.txt
```

Record the original SHA-256 hash:

Question 1. What security property are we testing?

Activity 3: Simulate Unauthorised Modification

Change the examination time:

```
sed -i.bak 's/Time: 09:00/Time: 14:00/' examination.txt
cat examination.txt
openssl dgst -sha256 examination.txt
```

Record the modified SHA-256 hash:

Question 2. Compare the two hashes. What do you observe?

Question 3. Could the Examinations Office detect the modification if it had securely obtained the original hash? Explain.

PART B - Protecting Confidential Student Information

Scenario: The Registrar's Office needs to transfer a fictional student record. Do not use real student information in this practical.

Activity 4: Create the Student Record

```
cat > student_record.txt << 'EOF'
Student Number: 202600001
Programme: BSc Computer Science
Final Mark: 78
Result: PASS
EOF
cat student_record.txt
```

Question 4. What could happen if an unauthorised person obtained this type of information?

Activity 5: Encrypt the Student Record

```
openssl enc -aes-256-cbc \
-salt \
-pbkdf2 \
```

```
-in student_record.txt \
-out student_record.enc
```

Enter a password when prompted, then examine the encrypted file:

```
cat student_record.enc
```

Can you read the original student information?

Activity 6: Decrypt the Record

```
openssl enc -aes-256-cbc \
-d \
-pbkdf2 \
-in student_record.enc \
-out student_record_decrypted.txt

cat student_record_decrypted.txt
diff student_record.txt student_record_decrypted.txt
```

Question 5. Which cryptographic security objective was achieved?

Question 6. What type of encryption did you use?

PART C - The Key-Distribution Problem

The Registrar and Faculty Administrator can encrypt information using AES, but they still need a secure way to establish the secret key or password. Sending that secret through ordinary email could expose it to an attacker.

PART D - Public-Key Cryptography

Scenario: The Examinations Officer must receive confidential information from academic departments. The officer creates a public/private key pair. The public key may be distributed; the private key must remain protected.

Activity 7: Generate the Examinations Officer's Private Key

```
openssl genpkey \
-algorithm RSA \
-out exams_private.pem \
-pkeyopt rsa_keygen_bits:2048

chmod 600 exams_private.pem
ls -l exams_private.pem
```

Question 7. Why should the private key never be emailed or placed in a publicly accessible folder?

Activity 8: Generate the Public Key

```
openssl pkey \
-in exams_private.pem \
-pubout \
-out exams_public.pem

cat exams_public.pem
```

Question 8. Which file can be distributed to University departments: exams_private.pem or exams_public.pem?

PART E - Sending Confidential Information to the Examinations Office

Activity 9: Create the Message

```
cat > lecturer_message.txt << 'EOF'
4CPS505 examination paper approved for printing.
Number of candidates: 120
EOF
```

Activity 10: Encrypt for the Examinations Officer

```
openssl pkeyutl \
-encrypt \
-pubin \
-inkey exams_public.pem \
-in lecturer_message.txt \
-out lecturer_message.enc \
-pkeyopt rsa_padding_mode:oaep
```

Activity 11: Decrypt at the Examinations Office

```
openssl pkeyutl \
-decrypt \
-inkey exams_private.pem \
-in lecturer_message.enc \
-out received_message.txt \
-pkeyopt rsa_padding_mode:oaep

cat received_message.txt
```

Question 9. Whose public key did the lecturer use?

Question 10. Which key was required to decrypt the information?

Stop and Think

Step	Key / Operation
Lecturer	Encrypts using: _____
Network	Carries ciphertext

Step	Key / Operation
Examinations Officer	Decrypts using: _____
Security objective	_____

PART F - Authenticating an Official University Notice

Scenario: Students receive a message claiming that a CSC301 examination has been cancelled. They need a way to verify that an official notice really came from an authorised source and was not modified.

Activity 12: Generate the HOD's Keys

```
openssl genpkey \
  -algorithm RSA \
  -out hod_private.pem \
  -pkeyopt rsa_keygen_bits:2048

openssl pkey \
  -in hod_private.pem \
  -pubout \
  -out hod_public.pem
```

Activity 13: Create an Official Notice

```
cat > official_notice.txt << 'EOF'
OFFICIAL NOTICE

The CSC301 examination will take place as scheduled.

Head of Department
Department of Computer Science
EOF
```

Activity 14: Sign the Notice

```
openssl dgst \
  -sha256 \
  -sign hod_private.pem \
  -out official_notice.sig \
  official_notice.txt
```

Activity 15: Verify the Notice

```
openssl dgst \
  -sha256 \
  -verify hod_public.pem \
  -signature official_notice.sig \
  official_notice.txt
```

Expected result: Verified OK

Question 11. Which key created the signature?

Question 12. Which key verified the signature?

PART G - Simulating a False University Notice

```
sed -i.bak \
's/will take place as scheduled/has been cancelled/' \
official_notice.txt

cat official_notice.txt

openssl dgst \
-sha256 \
-verify hod_public.pem \
-signature official_notice.sig \
official_notice.txt
```

Question 13. Why does the signature no longer verify?

Question 14. What two important security properties does this demonstrate?

PART H - A More Sophisticated Attack

An attacker could create their own key pair and claim that the attacker's public key belongs to the HOD. A valid signature is only meaningful if the verifier can trust the public key used for verification. This leads to digital certificates and Public Key Infrastructure (PKI).

PART I - University Digital Certificate

Activity 16: Create a Demonstration Certificate

```
openssl req \
-new \
-x509 \
-key hod_private.pem \
-out university_certificate.pem \
-days 365 \
-subj "/C=ZA/O=Example University/OU=Computer Science/CN=cs.example.ac.za"

openssl x509 \
-in university_certificate.pem \
-text \
-noout
```

Record the following certificate information:

Certificate field	Observed value
Subject	
Issuer	

Certificate field	Observed value
Valid From	
Valid Until	
Public-Key Algorithm	
Signature Algorithm	

PART J - Certificate Authorities and PKI

A self-signed certificate is useful for this laboratory demonstration, but public websites normally use certificates that participate in a broader chain of trust: Trusted Root CA -> Intermediate CA -> University Certificate -> University Website.

Activity 17: Examine a Real University's HTTPS Certificate

```
openssl s_client \
  -connect www.example.ac.za:443 \
  -servername www.example.ac.za \
  -showcerts </dev/null
```

Replace www.example.ac.za with the hostname supplied by your lecturer. Identify the website certificate, issuer, certificate chain, validity period and verification result.

PART K - How HTTPS Uses Cryptography

When a student connects to a University service over HTTPS, public-key mechanisms help authenticate the server and establish secure session keys. Efficient symmetric encryption then protects the bulk of the communication.

Stage	What happens
1	Student's browser connects to the University server.
2	The server presents a digital certificate.
3	The browser checks the certificate, hostname and trust chain.
4	Cryptographic session keys are established.
5	Symmetric encryption protects the subsequent communication.

PART L - University Security Incident

For each incident below, identify the security requirement and an appropriate cryptographic mechanism.

Incident	Security Requirement	Cryptographic Mechanism
Someone reads confidential student results while they are being transferred.		
An examination paper is modified		

Incident	Security Requirement	Cryptographic Mechanism
after approval.		
A fake examination notice is distributed claiming to come from the HOD.		
An attacker provides a fake public key claiming it belongs to the University.		
A laptop containing sensitive University information is stolen.		
Students need to confirm that downloaded software has not been modified.		

PART M - Group Challenge: Secure Examination Workflow

Work in pairs. One student acts as the Lecturer and the other as the Examinations Officer.

Student A - Lecturer

- Generate lecturer_private.pem and lecturer_public.pem.
- Create a fictional examination approval message.
- Sign the message using the lecturer's private key.

Student B - Examinations Officer

- Generate exams_private.pem and exams_public.pem.
- Give only the public key to the Lecturer.
- Never share the private key.

Complete the Secure Exchange

- The Lecturer encrypts a short confidential message using the Examinations Officer's public key.
- The Lecturer sends the encrypted message, signature and lecturer public key.
- The Examinations Officer decrypts the message and verifies the lecturer's signature.
- Modify the signed message deliberately and attempt verification again.

PART N - Security Analysis

Question 15. Which mechanism protects confidentiality?

Question 16. Which mechanism provides evidence that the message has not been modified?

Question 17. Which mechanism helps establish who signed the message?

Question 18. Why would a PKI still be required in a real University implementation?

PART O - Critical Thinking

Question 19. A staff member says, "We encrypt student records, therefore we do not need access controls." Explain why this statement is incorrect.

Question 20. A lecturer emails their private key to another lecturer because both need to sign official departmental documents. Identify the security problem and state what should be done instead.

Question 21. A student successfully verifies a digitally signed examination notice. Can the student automatically conclude that it came from the University? Explain why trust in the public key is also necessary.

Question 22. Why would it normally be inefficient to encrypt a very large database entirely using RSA public-key encryption? What approach is commonly used instead?

PART P - Putting the Module Together

University Requirement	Technology / Mechanism
Protect student information from unauthorised disclosure	
Detect modification of a downloaded file	
Efficiently encrypt large quantities of data	
Allow someone to encrypt information for a particular recipient	
Verify an authorised person's digital signature	

University Requirement	Technology / Mechanism
Associate an identity with a public key	
Establish institutional/public trust in certificates	

Final Scenario

A student logs into the University's online learning platform over HTTPS and submits an assignment. Explain how cryptography contributes to the following:

Question F1. Confidentiality: Can another person easily read the student's login credentials while they travel across the network?

Question F2. Integrity: Can modification of protected communication be detected?

Question F3. Authenticity: How can the student's browser gain confidence that it is communicating with the intended University server?

Question F4. Trust: Why does the browser trust the University's digital certificate?

Practical Submission

Submit a Cryptography Practical Report containing:

- Your name and student number.
- Answers to all practical questions.
- The original and modified SHA-256 hashes.
- Evidence of successful AES encryption and decryption.
- Evidence of successful public-key encryption and decryption.
- Evidence of successful digital-signature verification.
- Evidence showing failed verification after modifying a signed document.
- The relevant details extracted from the digital certificate.
- Your completed University Security Incident table.
- A 200-250 word reflection explaining how cryptography can protect information within a University environment.

Security Warning: Do not include passwords or private keys in your report. Do not use real student records, examination papers, passwords, marks or other confidential University information during this practical.

Key Concepts

Security Goal / Concept	Main Mechanism / Meaning
Confidentiality	Encryption
Integrity	Cryptographic hashes, MACs and/or digital signatures
Authenticity	Digital signatures or other authentication mechanisms
Symmetric cryptography	Shared secret keying material; efficient bulk encryption
Public-key cryptography	Public/private key pairs; authentication and key establishment
Digital certificates	Bind identities to public keys
PKI	Provides a trust framework for certificates and public keys

A secure University information system combines cryptography with appropriate authentication, authorisation, access control, secure configuration, monitoring, policy and human responsibility.